NOVI INDUS

PROJECT DOCUMENT

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1. Introduction

1.1 Purpose

This document defines the requirements for an **Education Web Application** that offers **live and pre-recorded classes**, **MCQ exams**, and **study materials (notes)** for students. The system allows **faculties** to conduct and manage classes while **Super Admin** oversees the entire platform. The web app includes a **freemium access model** and **course purchasing options** for premium users.

1.2 Scope

The system provides:

- Live classes through **Zoom integration**.
- Pre-recorded classes hosted on **YouTube** (to save storage costs), embedded using a **custom external player** that masks YouTube branding.
- MCQ-based exams for self-assessment.
- Notes uploaded by faculties for each course.
- Role-based dashboards for **Faculty** and **Super Admin**.
- Course purchasing system for paid content.
- Freemium access for non-registered or free users to browse limited content.

1.3 Users

- **Student (Free & Paid)** attends classes, attempts exams, views notes.
- **Faculty** creates and manages courses, uploads content, conducts live classes, monitors performance.
- **Super Admin** full control over the platform, manages faculties, courses, payments, and reports.

2. Overall Description

2.1 Product Perspective

This is a web-based learning platform accessible via desktop and mobile browsers.
Key modules:

1.

1. Student Portal (free and paid access)
2. Faculty Dashboard
3. Super Admin Dashboard
4. Integrated Zoom for live classes
5. YouTube video integration for pre-recorded content
6. MCQ test engine
7. Notes & resources management
8. Payment gateway for course purchase

2.2 Operating Environment

- **Frontend:** React.js / Next.js
- **Backend:** Python (REST APIs)
- **Database:** MySQL / PostgreSQL
- **Hosting:** Cloud-based (AWS / DigitalOcean)
- **Video hosting:** YouTube API integration (videos unlisted, embedded in custom player)
- **Live class:** Zoom API (OAuth-based integration)
- **Payment:** Razorpay(as per client region)

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3.1.2 Product Listing Page (PLP)

- Filter by Category, Brand, Scent Family, Price Range.
- Sort by Popularity, New, Price Asc/Desc.
- “Quick View” option with Add to Cart.
- Wishlist & Compare options.

3.1.3 Product Detail Page (PDP)

- Product images, descriptions, scent notes, and variants (size/type).
- Display of **base price** and **premium perfume surcharge (+20 AED)**.
- Related products & “Frequently Bought Together.”
- Customer reviews and star ratings.
- Real-time stock & price display.
- Add to Cart via AJAX (no page reload).

3.1.4 Custom Box Builder

- Users can **create a box with 3 – 6 perfumes**.
- Each box has a **base price** (admin configurable).
- Certain products (premium perfumes) add **+20 AED per item**.
- Dynamic total calculation via AJAX.
- Visual box-building interface (drag-and-drop or selection grid).
- Admin can define eligible perfumes, limits, and conditional pricing.
- If the same perfume is selected twice, rules (like “+50 AED after second”) can apply.

3.1.5 Cart & Checkout

- AJAX cart sidebar (live updates).
- Quantity update and coupon apply without reloading.
- One-page checkout (address + payment + review).
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- # 3. Functional Requirements
- ## 3.1 User Management
- **FR01** – Registration for Students (email, phone, password, verification).
  FR02 – Faculty and Admin accounts created by Super Admin.
  FR03 – Forgot Password via email/OTP.
  FR04 – Freemium users can browse courses but cannot access premium content fully.

3.3 Live Classes (Zoom Integration)

- **FR09** – Faculty schedules live classes via Zoom API integration.
  FR10 – Students receive notification and link to join class directly from web app.
  FR11 – Attendance automatically recorded (based on Zoom participant list).
  FR12 – After completion, class details are stored in course history.

3.4 Pre-Recorded Classes (YouTube Integration)

- **FR13** – Faculty uploads videos to YouTube (unlisted).
  FR14 – System stores YouTube video ID and plays it in an **external player** that hides YouTube branding and links.
  FR15 – Students can view videos based on access level (free or paid).

3.5 Notes & Study Material

- **FR16** – Faculties can upload notes (PDF, DOCX, PPT, images).
  FR17 – Students can download or view notes under each chapter.
  FR18 – Super Admin can monitor and edit uploaded notes.

3.6 MCQ-Based Exams

- **FR19** – Faculties can create MCQ exams for courses.
  FR20 – Randomized questions from database during each attempt.
  FR21 – Students get instant score and correct answers after submission.
  FR22 – Admin can view exam statistics per course/student.

3.7 Payments & Course Purchase

- **FR23** – Students can buy full courses using integrated payment gateway.
  FR24 – After successful payment, course is unlocked.
  FR25 – Super Admin can manage pricing, offers, and discounts.
  FR26 – Generate invoices for payments.

3.8 Freemium Access

- **FR27** – Free users can view limited preview lessons or demo courses.
  FR28 – Premium content locked with a “Buy Course” prompt.
  FR29 – Super Admin can control which lessons are free.
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3.9 Dashboards

- **Faculty Dashboard:**
  - Manage own courses, upload materials, schedule live classes, track student engagement.
- **Super Admin Dashboard:**
  - Manage faculties, students, payments, reports, and full system control.

3.10 Notifications & Communication

FR30 – Email/SMS notifications for class schedule, purchases, exam results.
FR31 – In-app notifications for live class reminders.
FR32 – Automated mail after successful course purchase.

5. External Interfaces

5.1 Zoom API

- Create/schedule/join meetings using faculty credentials.
- Store meeting IDs, links, and attendance reports.

5.2 YouTube API

- Fetch and embed video using unlisted video IDs.
- Custom player built to hide YouTube controls and branding.

5.3 Payment Gateway

- Handle one-time course purchases.
- Secure callbacks to verify payment success/failure.

5.4 Notification Services

- Email via SendGrid/Mailgun.
- SMS via Twilio/local provider.
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6. Data Model (High Level)

- **User (Student/Faculty/Admin)**
- **Course**
- **Lesson (Live/Recorded)**
- **Exam (MCQ)**
- **QuestionBank**
- **Notes**
- **PaymentTransaction**
- **Notification**

7. Use Cases

UC01: Attend Live Class

1. Faculty schedules live class via dashboard.
2. Zoom meeting created automatically.
3. Students receive notification and join via embedded link.
4. Attendance logged automatically.

UC02: View Recorded Class

1. Student selects course → recorded lesson.
2. YouTube video ID fetched.
3. External player plays video (YouTube branding masked).

UC03: Take MCQ Exam

1. Student selects course exam.
2. Questions randomly loaded from question bank.
3. On submission, results displayed instantly.
1.

UC04: Buy Course

1. Student clicks “Buy Course”.
2. Redirected to payment gateway.
3. On success, course unlocked and receipt emailed.

KICK START

Noviindus Technologies LLP will kick start the application development only after receiving the LPO.

PROPOSAL VALIDITY

15 Days

Contact us

You can get in touch with us in any of the below ways:

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If you would like to proceed with our proposal then you can sign the first page and return a copy to us by email or post.

In any case please feel free to call us to discuss the quote, request more information or for any other reason.

We look forward to hearing from you soon!